Mohamad Maulana Firdaus Ramadhan

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Location: Bandung, Indonesia

Education

- **Bandung Institute of Technology (ITB)** — Bandung, Indonesia
  Bachelor of Information System and Technology — *June 2022 - 2026 (Expected)*
  Relevant Coursework: *Artificial Intelligence Fundamentals, Natural Language Processing (NLP), Speech Recognition, Data Mining, Data Analytics, Object-Oriented Programming, Probability and Statistics, Database Modeling.*

Skills Summary

Languages	Python, SQL, C++, Shell Scripting
AI & GenAI	LLM, Agentic RAG (LangChain), Transformer Architecture, LoRA, Quantization (GPTQ)
Cloud	AWS (EC2, S3, ECS, SQS, EventBridge), Kubernetes (K8s), Docker, CI/CD
Libraries	Hugging Face, PyTorch, LangGraph, Scikit-learn, Pandas, NumPy, OpenCV
Databases	FAISS, Chroma, Pinecone, Qdrant, PostgreSQL, Redis (Queues)
Soft Skills	Cross-functional Collaboration, Technical Leadership, Problem Solving

Experience

- **Neuram** — Jakarta, Indonesia
  AI Software Engineer — *Feb 2026 – April 2026*
  - **Voice AI Infrastructure**: Architected an automated appointment system by integrating SIP INVITE protocols from client domains to Neuram's Twilio infrastructure and ElevenLabs webhooks.
  - **Dynamic Context Injection**: Developed backend endpoints to fetch real-time client data (customer profiles, available slots) and inject them as dynamic variables into ElevenLabs agents for personalized calls.
  - **End-to-End Workflow Automation**: Implemented automated post-call processing by triggering API callbacks to client systems, delivering call outcomes and data insights immediately after session termination.
  - **Advanced Agent Capabilities**: Leveraged ElevenLabs server-side tools to extend agent functionality, enabling complex logic execution and tool-calling during real-time voice interactions.
- **Legali** — Bandung, Indonesia (Remote)
  AI Software Engineer — *Feb 2026*
  - **Smart Legal Drafter Improvement**: Enhanced drafting workflows by implementing conditional logic for template-based vs. custom generation, tailored to user profiles (legal vs. non-legal backgrounds).
  - **Litigation Case Builder & State Management**: Refined the internal state machines of the litigation case builder to ensure robust data flow and improved accuracy in tracking case progress.
  - **Ecosystem Expansion**: Developed Google Docs and Microsoft Word extensions to integrate AI legal tools directly into professional writing environments, increasing user productivity.
  - **Personalization & Prompt Engineering**: Engineered a "Prompt Vault" for pre-configured legal templates (e.g., NDA generation) and built a "Concierge" system to deliver personalized AI responses based on user-specific data.
- **AI Center ITB** — Bandung, Indonesia
  AI Research Assistant — *Oct 2025 - Dec 2025*
  - **Agentic RAG System**: Engineered an enterprise-grade Agentic RAG system for the Oil & Gas sector, enabling complex QnA and high-precision data extraction from unstructured documents.
  - **Hybrid Search Implementation**: Orchestrated Qdrant Hybrid Search (Dense + Sparse) to maximize retrieval precision for domain-specific queries, solving low-recall issues in technical documentation.
  - **Text-to-SQL Pipeline**: Constructed a dynamic Text-to-SQL workflow using LangGraph to facilitate seamless interaction with structured databases, allowing users to query SQL data using natural language.
  - **LLM Evaluation Harness**: Established an automated "LLM-as-a-Judge" harness to validate model performance across 1,000+ samples, ensuring production readiness and reducing hallucination rates.
- **Bandung Institute of Technology** — Bandung, Indonesia
  AI Research Assistant — *March 2025 - Dec 2025*
  - **Deep Learning Architecture**: Partnered with faculty to engineer a real-time AI system for partial discharge signal classification, achieving 100% accuracy using LSTM with Attention mechanisms and Dense Concatenation.
  - **Edge Computing Deployment**: Deployed optimized signal processing algorithms onto FPGA and Raspberry Pi 5, ensuring efficient real-time data collection and inference at the edge.
  - **Data Acquisition & Processing**: Collected and processed a robust dataset of 2,312 signal samples using Analog Discovery 2 for high-fidelity data analysis.
  - **Monitoring Dashboard**: Spearheaded the development of a web-based monitoring dashboard to visualize signal predictions and track system status in real-time.
- **PT. Solutopia Untuk Nusantara (SPUN)** — Jakarta, Indonesia
  AI Engineer Intern — *July 2025 - Sept 2025*
  - **Production AI Assistant**: Architected a production-ready AI assistant for visa services, handling multi-turn conversations and integrating model optimization techniques for efficient inference.

- **Cloud Infrastructure & Scalability**: Designed and deployed scalable AI services on AWS using ECS and EC2, utilizing Amazon S3 for secure document storage and Secrets Manager (KeyStorage) for credential management.
- **Event-Driven Architecture**: Implemented an asynchronous processing pipeline using Amazon SQS and EventBridge to handle high-concurrency requests, ensuring system reliability and fault tolerance.
- **CI/CD Automation**: Automated the deployment lifecycle using AWS CodePipeline, reducing deployment time and ensuring consistent delivery of AI-enabled tools into the production environment.
- **Multi-Source RAG Pipeline**: Engineered an intelligent RAG pipeline using Pinecone and PostgreSQL.

Projects

- **ChatDoctor** [Code]: Earned Top-10 National Finalist. Developed a healthcare chatbot powered by Large Language Models (LLM) to enhance medical accessibility. Designed the system to provide accurate, context-aware medical information. Tech: LLM, Python.
- **BattleTalk** [Code]: Top 10 Finalist FindIT UGM. Built an AI-powered English learning application featuring gamification to enhance user engagement. Tech: Vite, Qwen2.5, Supabase, Ollama, Node.js.
- **Trash Reporting System** [Paper]: National Finalist GEMASTIK XVII. Created an AI-powered solution for automated trash reporting. Trained custom YOLOv8 models for real-time object detection in various environments. Tech: YOLOv8, Computer Vision, Python.
- **RoadEye** [Code]: Top 20 Teams GDSC Hackfest. Built a traffic monitoring system utilizing AI for anomaly detection and traffic density analysis. Tech: Next.js, Python, PostgreSQL, AI.
- **Marketplace Recommender System** [Presentation]: 1st Place Hology 7.0. Optimized user/seller experience in TokoGenZ through clustering, recommender systems, and interactive Tableau dashboards. Tech: K-Means Clustering, Tableau, Python.

Leadership & Organization

- **Data Community HMIF ITB** — Bandung, Indonesia
  Head of Community — *June 2024 - Present*
  - Directed strategic initiatives and activities related to data science and machine learning for the student body. Coordinated technical workshops and events designed to deepen members' proficiency in data engineering and analytics.
- **Google Developer Student Club ITB** — Bandung, Indonesia
  Curriculum Developer — *Sept 2023 - Aug 2024*
  - Designed and developed a comprehensive Machine Learning curriculum, including structured roadmaps and hands-on tasks for 800+ members. Facilitated project-based learning assignments to enhance practical skills.
- **STEI-K Student Council** — Bandung, Indonesia
  Database Division Staff — *Nov 2022 - Oct 2023*
  - Managed and processed student data to create actionable information. Generated analytics reports using Python Pandas and Excel. Awarded Best Staff of the Month (Feb 2023).

Invited Speaker

- **Symphony of Data: The Key to Brilliant Decision Making** [Certificate] — KAT ITB, 2024
  Guest Speaker
  - Presented on the strategic use of data analytics for decision-making processes at a centralized training event.
- **Data Competition Strategies** [Certificate] — Kaderisasi Wilayah STEI-K ITB 2023, 2024
  Guest Speaker
  - Spoke on innovations in data analysis and machine learning development for student competitions.
- **Introduction to Data Science** — SPARTA HMIF 2023, 2024
  Guest Speaker
  - Delivered an introductory session on data science concepts and career paths for new students.

Certifications

- **Data Scientist Professional Certificate** [Certificate]: Certified proficiency in Data Management, Exploratory Analysis, Statistics, Model Development, and Production Coding.
- **Data Analyst Professional Certificate** [Certificate]: Certified competencies in Analytic Fundamentals, Visualization, and Reporting techniques.
- **Kalbe Nutritionals Data Scientist Internship** [Certificate]: Awarded Certificate of Achievement for successful Machine Learning implementation and Business Acumen application.
- **Data Analysis Certification (FreeCodeCamp)** [Certificate]: Hands-on validation of Python data libraries including NumPy, Pandas, Matplotlib, and Seaborn.
- **Machine Learning with Python (FreeCodeCamp)** [Certificate]: Completed comprehensive coursework on implementing ML algorithms with Python.

Honors and Awards

- **1st Place, Hology 7.0 Data Mining (National)** [Certificate]: Outperformed 235 teams by developing a marketplace optimization solution using K-Means Clustering and Tableau dashboards.
- **National Finalist (Top 20), GEMASTIK XVII Data Mining** [Certificate]: Developed an AI-powered Trash Reporting and Detection System using custom-trained YOLOv8 models for real-time inference.
- **Top 10 Finalist, Data Science National Tournament** [Certificate]: Created "ChatDoctor," a healthcare chatbot leveraging Large Language Models (LLM) to democratize access to medical information.
- **Top 10 Finalist, Objective Quest Unair (Airnology 3.0)** [Certificate]: Selected as a finalist among 100+ teams by proposing an innovative data-driven solution for air quality monitoring.
- **Top 10 Finalist, FindIT Data Analytics Competition UGM** [Certificate]: Competed against 200+ teams to solve complex business problems using advanced data analytics and predictive modeling.
- **Top 10 Finalist, FindIT UGM Hackathon - 2025** [Certificate]: Engineered an AI-powered English learning app with gamification using Vite, Qwen2.5, and Supabase.
- **Top 10 Finalist, Ristek Datathon UI - 2025** [Certificate]: Selected as a top finalist among 200+ participants, solving data challenges using machine learning optimization.
- **National Finalist (Top 20), GEMASTIK XVI Data Mining** [Certificate]: Developed "Automated Traffic Condition Monitoring," a computer vision system using YOLOv8 for real-time traffic anomaly detection.
- **Top 20 Teams, GDSC Hackfest Indonesia** [Certificate]: Built "RoadEye," a comprehensive traffic monitoring solution integrated with Next.js and Python-based AI analytics.